MIL-D-435F(ME)
AMENDMENT 4
19 July 1988
SUPERSEDING
AMENDMENT 3
1 July 1987

MILITARY SPECIFICATION

DISTRIBUTORS, LIQUID BITUMINOUS MATERIAL,

TANKLESS-TYPE, TRAILER-MOUNTED

This amendment forms a part of MIL-D-435F(ME), dated 20 August 1985 and is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1, under SPECIFICATIONS, MILITARY, add:

| | | |
|---|---|---|
| * | "MIL-C-1283 | - Can, Gasoline, Military, 5 Gallon. |
| * | MIL-P-14105 | - Paint, Heat Resisting, (for Steel Surfaces) |
| * | MIL-T-83133 | - Turbine Fuel, Aviation Kerosene Type, Grade JP-8." |

2.1.1, under STANDARDS, MILITARY, add:

| | | |
|---|---|---|
| * | "MIL-STD-1791 | - Designing for Internal Aerial Delivery in Fixed Wing Aircraft. |
| | MS52131 | - Connector, Plug, Electrical Intervehicle Power Cable. |
| * | MS53052 | - Bracket Assembly, Liquid Container, Five Gallon." |

2.2, add:

"ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Department of Defense Materiel on Open Top Cars, section 6.

(Application for copies should be addressed to the Association of American Railroads, 50 F Street, Washington, DC 20001.)"

PAGE 5

3.1, line 1, delete "gasoline or".

AMSC N/A FSC 3895

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:26Z
Check the source to verify that this is the current version before use.

PAGE 6

3.2, line 1, delete “When specified” and substitute “Unless otherwise specified”.

PAGE 7

3.4, line 3, add the following sentence:

“Also, the trailer shall allow the distributor to be transportable by rail and water modes and in C-130, C-141, and C-5 aircraft.”

PAGE 13

TABLE II, under Lamp lenses and bulbs, column 2, add “Lamp lenses and bulbs shall be

PAGE 14

After 3.4.15.1, add the following:

“3.4.15.2 Slaving components. The distributor shall be equipped with a 24 volt slave receptacle conforming to MS52131. The slave receptacle shall permit charging of the batteries and slave starting of the engine from an external energy source and shall also provide a power source for charging and slaving other equipment. The slave receptacle shall be installed on the exterior of the distributor near the battery compartment and shall be accessible to personnel standing on the ground. The voltage potential of the salve receptacle shall be identified on a plate mounted adjacent to the receptacle. The plate shall be in accordance with the requirements of 3.17.1. (This slave receptacle shall be compatible with NATO slave cable NSN 2590-00-148-7961.)”

3.5, line 1, delete “gasoline or”.

After line 13, add the following:

“e. Oil sampling valve.”

PAGE 15

* 3.5.3, delete in its entirety and substitute:

“3.5.3 Engine fuel. The diesel engine furnished shall operate on fuel conforming to both VV-F-800, grade DF 2 and MIL-T-83133, grade JP-8.”

* “3.5.6, line 6, delete “(specify type of fuel)” and substitute “DIESEL OR JP-8”.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:26Z
Check the source to verify that this is the current version before use.

PAGE 20

* 3.8.4, line 2, after “conforming to” add “both MIL-T-83133, grade JP-8 and”. Line 5, delete “FILL WITH DIESEL FUEL ONLY” and substitute “FILL WITH DIESEL OR JP-8 FUEL

* 3.8.5, line 5, delete “FILL WITH FUEL ONLY” and substitute “FILL WITH DIESEL OR

PAGE 23

* 3.16, LINE 4, after “in” add “MIL-STD-1791, for equipment restraint criteria and”.

PAGE 24

3.18, 3.19 and 3.19.1, delete in their entirety and substitute:

“3.18 Treatment and painting. The portions of the distributor normally painted, internally and external, shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), top coat color shall be camouflage green 383 conforming to MIL-C-46168. When specified (see 6.2), the distributor shall be overcoated in accordance with the Government furnished camouflage patterns with MIL-C-46168 top coat of the colors specified in the camouflage patterns except the first article may be camouflage green 383. Heat resistant paint conforming to MIL-P-14105 shall be used for areas exceeding 400 ºF.

“3.19 Camouflage pattern data. When specified (see 6.2), the contractor shall provide camouflage pattern data.

PAGE 26

3.26, delete in its entirety and substitute:

“3.26 Government furnished property. When specified (see 6.2.1), the following property in the quantity indicated will be furnished by the Government (see 6.6).

| Item No. | Description | Quantity |
|---|---|---|
| 1 | Camouflage paint pattern | As required” |

After 3.26, add:

* “3.27 Rail transport. Distributor, and all attached equipment in rail transport configuration must be able to withstand transport by rail without damage. Overall



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:26Z
Check the source to verify that this is the current version before use.

dimensions shall be within the Gabarit International De Chargement (GIC) clearance diagram (see figure 2)."

* "3.28 Decontamination unit. A bracket assembly conforming to MS53052 shall be mounted by the contractor. The location of the bracket shall be such that it is accessible to personnel and provides clearance to remove and emplace a 5 gallon container conforming to MIL-C-1283. The location of the bracket assembly shall be away from heat producing sources."

PAGE 27

4.3.2, line 2, delete "4.5.13" and substitute "4.5.2.14 and 4.5.2.16".

PAGE 28

4.5.1, delete 152 and 153 and substitute:

"152. Camouflage pattern data not provided when specified (see 3.19)."

Add:

"164. Slave components not as specified (see 3.4.15.2).
* 165. Decontamination unit bracket not as specified (see 3.28)."

PAGE 29

* 4.5.2.1, line 6, add: "All tests as specified in 4.5.2.2 through 4.5.2.14 shall be run on MIL-T-83133, grade JP-8. All tests as specified in 4.5.2.15 shall be run on either VV-F-800, grade DF 2 or MIL-T-83133, grade JP-8."

4.5.2.5, delete the title and substitute: "Equipment tiedown attachments."

PAGE 30

* 4.5.2.5.1, line 3, after "in", insert: "MIL-STD-1791 and".

* 4.5.2.5.2, line 1, after "of", insert: "MIL-STD-1791".

PAGE 34

After 4.5.2.15.1, add:

"4.5.2.16 Rail impact test. Unless otherwise specified (see 6.2), the distributor and all attached equipment in rail transport configuration shall be tested in accordance with the following rail impact test procedure:

STEP 1. From one to five rail cars shall be positioned on a level section of track. The following conditions shall be satisfied:



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:26Z
Check the source to verify that this is the current version before use.

a. The total weight of the car(s) shall be at least 114,000 kilograms (250,000 pounds).
b. The couplers between cars shall be compressed to take up any slack.
c. All of the air and hand brakes on the car(s) used shall be set.
d. Any load in or on the car(s) shall be secured to prevent sliding or shifting; any movement greater than 5 cm shall be justification for retest.
e. The end of the buffer car to be struck must have a standard draft gear.

STEP 2. Mount the test item on the test car. The test car shall be equipped with standard draft gears and conventional underframes. Mounting of the test item shall incorporate the standard loading and bracing method as shown in section 6 of the Association of American Railroads (AAR) "Rules Governing the Loading of Department of Defense Materiel on Open Top Cars". No exotic or unusual tiedown methods shall be used; any non-standard loading and bracing must be approved by the Military Traffic Management Command Transportation Engineering Agency (MTMCTEA), ATTN: MIT-TR, P.O. Box 6276, Newport News, VA 23606-0276 prior to testing.

STEP 3. Situate the test car between the buffer cars and the locomotive, and pull the test car at least 65 meters (200 feet) from the buffer car(s) along a level section of track (a minimum distance to achieve the required locomotive speeds).

STEP 4. Position the knuckles of the buffer and test cars for coupling.

STEP 5. Install a timing device to measure the test car speed (±.1 km/hr) just prior to impact with the buffer car(s). Suggested methods include electronic timing (microswithes) and radar. The use of torpedoes and a stopwatch is permissible but not recommended because of the inaccuracies involved.

STEP 6. Push the test car toward the buffer car(s) and, by using the locomotive's speedometer or other means, release the test car when the desired test speed is reached, thus allowing the test car to freely impact the buffer car(s).

STEP 7. Repeat step 6 until the test car(s) has impacted the buffer car three times at the same end, once each at speeds of 6.4, 9.7 and 13.0 km/hr (4, 6, and 8 mph) +0.8, -0.0 km/hr. Reverse the test car and repeat the 13.0 km/hr impact, for a total of 4 impacts.

NOTE. Adjustment of the lading or securing mechanism, or reconditioning of the bracing or items of securement is not allowed with a complete retest.

"4.5.2.16.1 Failure criteria. Failure to meet the requirements of the AAR rail impact test and the requirements specified in 3.27 shall constitute failure of this test."



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:26Z
Check the source to verify that this is the current version before use.

6.2q, delete and substitute:

"q. When camouflage patterns will be provided (see 3.

Add:

"u. When rail impact test is not required (see 4.5.2.16)."

6.2.2, delete and substitute:

"6.2.2 Data requirements. When this specification is used for an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

| Paragraph No. | Data requirements | Applicable DID No. |
|---|---|---|
| a. 3. | Color photograph prints | DI-MISC-80192 |
| b. 3. | Camouflage line art data | DI-MISC-80176 |

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

Figures 2 and 3, delete entirely and substitute new figure 2.

The margins of the amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletion) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army - ME

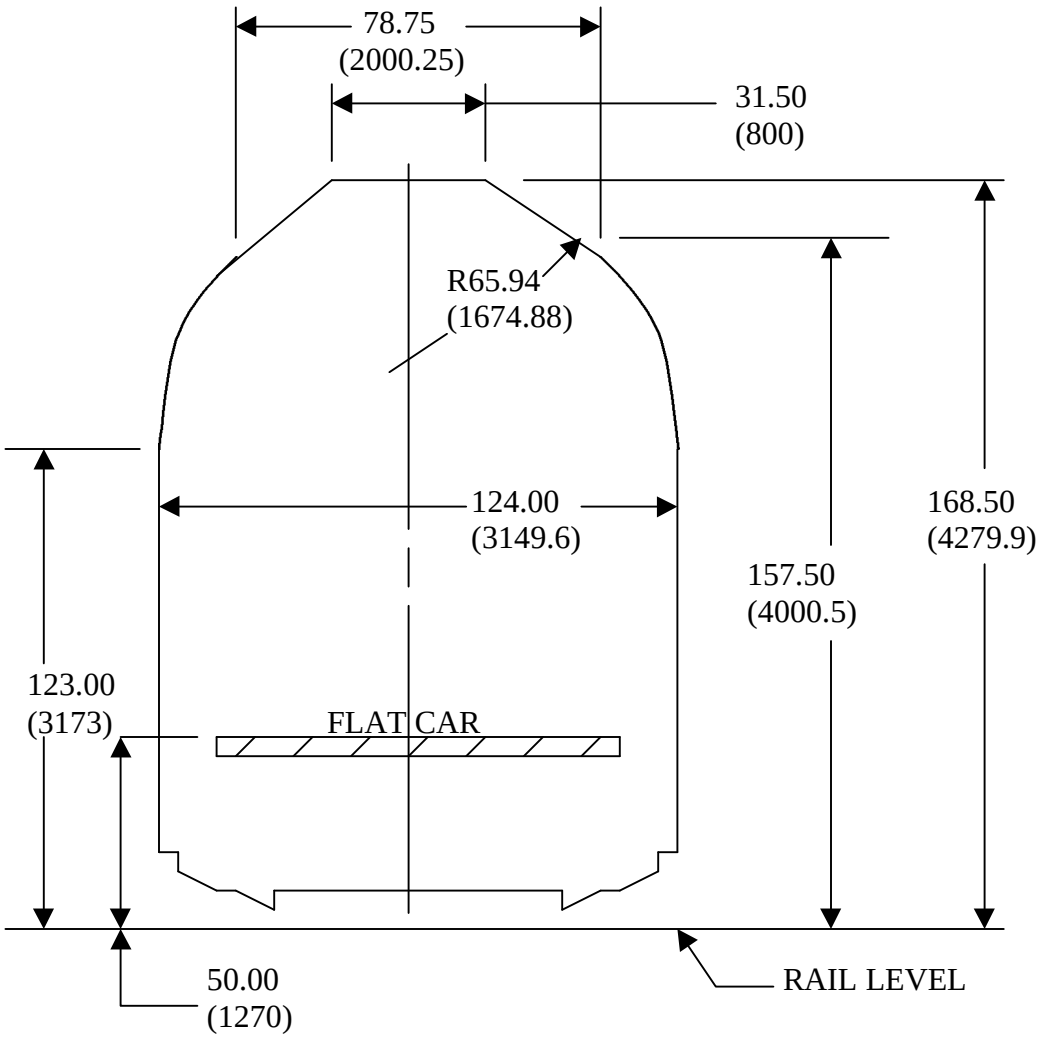
Preparing activity:
Army - ME

Review activity:
Army - AT

Project 3895-A273



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:26Z
Check the source to verify that this is the current version before use.




NOTE: DIMENSIONS SHOWN: INCHES (MILLIMETERS)

FIGURE 2. Gabarit International De Chargement (GIC) clearance diagram.

X-4093C



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:26Z
Check the source to verify that this is the current version before use.